

backup stores part of the energy that otherwise goes unutilised. There is a tradeoff between the amount of energy to be backed up and the overall cost of the system. A partial backup offers improved utilisation of SDL while keeping the system's cost low. (However, this proposed design cannot be used to charge the battery through AC mains.)

The energy thus stored can be used during low sunlight, cloudy weather, and also in the evenings. The energy stored during weekends can be utilised during weekdays. This circuit can also be used as an emergency lamp. The author's prototype is shown in Fig. 1.

PV and battery voltage matching

PV panels and batteries follow different voltage standards. A 12V PV panel



Fig. 3: Three LED arrays placed in three aluminium T channels

Box 2: Designing LED arrays

$$\text{Number of LEDs} = V_{mp} / V_f = 35/3 = 11.6$$

Therefore maximum number of LEDs in series = 11

$$\text{Residual voltage} = V_{mp} - (11 \times V_f) = 35 - 33 = 2V$$

Max current for 1W LED is 330mA. Here, we will design for an LED current of 200mA.

$$\text{Value of current limiting resistance} = \text{Residual voltage} / I_{LED} = 2/0.2 = 10\Omega$$

$$\text{Number of LED arrays} = I_{mp} / I_{LED} = 0.58/0.2 = 2.9$$

Thus, we can have three LED arrays connected in parallel.

generates 17.5V at the maximum power point, while a lead-acid battery is designed for 12V nominal voltage. Following a combination of PV panels and batteries provides a good match.

If two 12V PV panels are connected in series, the total PV voltage generated at maximum power point (V_{mp}) = $17.5 \times 2 = 35V$.

And with three 12V lead-acid batteries connected in series, the nominal voltage of the battery bank (V_b) = $12 \times 3 = 36V$

Thus, we can see that the nominal voltage of a battery bank and the voltage of PV panels match very closely. There will be a slight mismatch in the voltages based on the state of charge of the battery, which is taken care of during circuit design.

The above combination is given as an example. Users can explore other combinations based on their power needs and investments. Power can be determined from the specifications of PV panels and batteries, as shown in Box 1.

Circuit and working

Circuit diagram of the solar day lamp with battery backup is shown in Fig. 2. It is built around two P-channel MOSFET IRF9540 (IRF1 and IRF2), BC556 transistor (T1), three BC546 transistors (T2-T4), two 10W photovoltaic panels (PV1 and PV2), three

PARTS LIST

Semiconductors:

IRF1, IRF2	- IRF9540 P-channel power MOSFET
T1	- BC556 transistor
T2-T4	- BC546 transistor
ZD1, ZD2	- 15V Zener diode
ZD3	- 36V Zener diode
D1	- SR5100 diode
D2-D7	- 1N4148 signal diode
LED1-LED33	- 33 LED arrays of 1-watt each
LED34	- 5mm LED (white)
LED35	- 5mm LED (blue)

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

R1-R3	- 10-ohm, 1-watt
R4	- 7.5-ohm, 5-watt
R5, R6, R9, R16	- 10-kilo-ohm
R7, R18	- 1-kilo-ohm
R8, R11, R13	- 22-kilo-ohm
R10	- 1-mega-ohm
R12	- 33-kilo-ohm
R14, R17	- 27-kilo-ohm
R15	- 47-kilo-ohm
VR1	- 1-kilo-ohm potmeter

Capacitors:

C1	- 10 μ F, 63V electrolytic
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Miscellaneous:

F1	- 2A fuse with holder
PV1, PV2	- 10-watt photovoltaic panel
BATT.1-BATT.3	- 12V, 7Ah lead-acid battery
S1, S2	- Double-pole double-throw (DPDT) switch
	- Heat-sink for IRF9540 MOSFET (2 number)
	- 2-pin terminal connector (3 number)

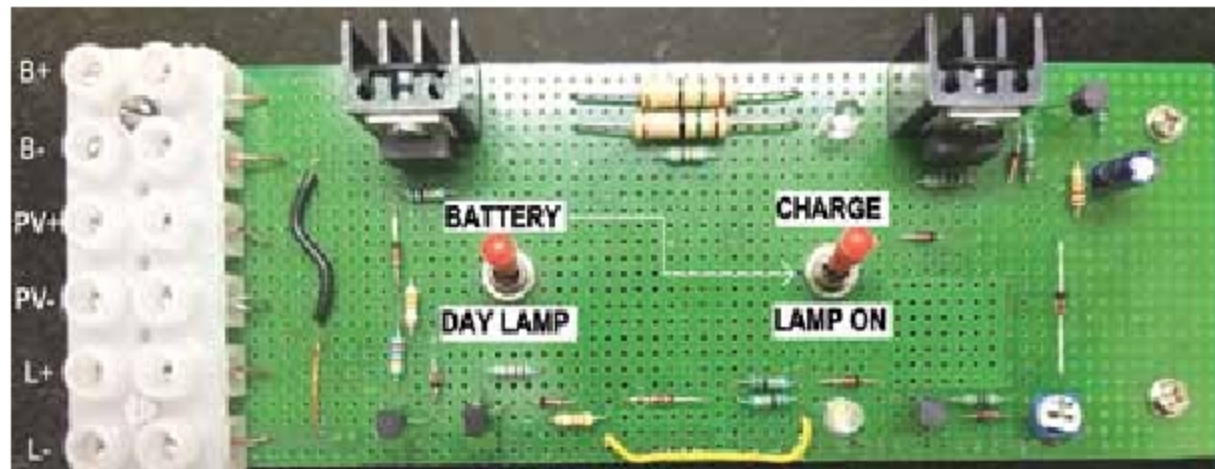


Fig. 4: Author's prototype showing PCB with different sections

12V, 7Ah batteries (BATT.1 through BATT.3), three white LED arrays (LED1 through LED33), 1-kilo-ohm potmeter (VR1), two DPDT switches (S1 and S2), and a few other components.

Modes of operations. The lamp consists of three LED arrays (L1, L2, and L3), which are placed in three aluminium T channels (refer Fig. 3). The three LED arrays are connected in parallel with a resistor in each array. Note that the values of resistors R1, R2, R3, and R4 given in this design are just examples. Based on the cable lengths used in the system, these values may have to